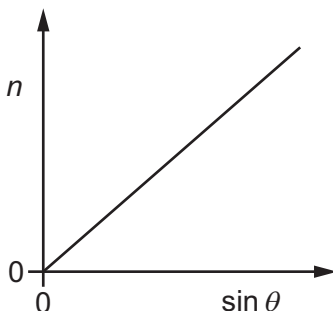


- 29 A student uses a diffraction grating to determine the wavelength of visible light from a source.

The diffraction grating has 300 lines per mm. The student measures the angle  $\theta$  of each order  $n$  of the intensity maxima. A graph of  $n$  against  $\sin \theta$  is plotted. The line of best fit for the plotted points is shown and has gradient  $G$ .



Which expression represents the wavelength, in m, of the visible light in terms of  $G$ ?

- A  $3 \times 10^5 G$       B  $3.3 \times 10^{-6} G$       C  $\frac{3.3 \times 10^{-6}}{G}$       D  $\frac{300}{G}$

- 30 Two waves meet.

What is **not** a necessary condition for the waves to produce a stationary wave?

- A They must be of the same type.  
B They must have the same period.  
C They must have the same wavelength.  
D They must travel in the same direction.

- 31 Light of a single wavelength is incident normally on a double slit. Interference fringes are observed on a screen.

The distance from the double slit to the screen is 0.60 m and the fringe separation is 1.8 mm.

The distance from the double slit to the screen increases by 0.90 m.

What is the new fringe separation?

- A 1.2 mm      B 2.7 mm      C 4.5 mm      D 5.4 mm

- 32 A wire is made from a metal of constant resistivity. There is a constant current in the wire.

Which statement about the potential difference across the wire is correct?

- A It is directly proportional to the length of the wire.  
B It is inversely proportional to the length of the wire.  
C It is directly proportional to the diameter of the wire.  
D It is inversely proportional to the diameter of the wire.